

Student Performance Analytics

Academic Outcomes, Equity Factors & Preparation Impact — an Excel/Python EDA and Power BI Semantic Model Case Study

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01 Executive Summary

This report analyses academic outcomes for 11,040 students using an Excel/Python exploratory data analysis workflow paired with a Power BI semantic model. The dataset records Math, Reading, and Writing scores alongside demographic, household, and preparation variables — gender, ethnic group, parental education, lunch programme enrolment, test-preparation completion, study hours, sport participation, and household composition.

The purpose of the analysis is to identify which factors are most strongly associated with student achievement, quantify the size of each effect, and separate levers that stakeholders (schools, counsellors, and programme administrators) can act on from those that are simply descriptive.

Overall performance

Average scores sit in the mid-60s to high-60s range across all three subjects — Math 66.7, Reading 69.4, Writing 68.4 — with a standard deviation of roughly 15 points in each, indicating a wide spread rather than a tightly clustered student population.

Most important findings

- the widest achievement gap in the dataset (nearly 12 points in Math) sits between students on standard vs. free/reduced lunch — the closest available proxy for household economic access;
- completing the test-preparation course is associated with a 4–9 point lift across subjects, yet only 34.1% of students with a recorded status have completed it;
- parental education level shows a near-monotonic, roughly 11-point gradient from "some high school" to "master's degree" households;
- gender differences run in opposite directions by subject — male students lead in Math, female students lead in Reading and Writing — so no single subject is uniformly stronger for either gender;
- household composition (marital status, first-child status, transport method) shows negligible effect (under 3 points) on scores, and should not be treated as a targeting variable.

Key business recommendations

- expand test-preparation access and track completion as a KPI, given its measurable lift and low current uptake;
- prioritise support resources (tutoring, materials, meal-adjacent support) for free/reduced-lunch students, where the largest gap is concentrated;
- treat literacy-focused interventions (Reading/Writing) as likely to reinforce each other, given their 0.95 correlation, while resourcing Math support separately;
- avoid using demographic categories such as ethnic group as a targeting lever on their own — flag them for monitoring, since they likely proxy unmeasured socio-economic conditions rather than causal drivers.

02 Business Problem

Schools and education programmes routinely collect demographic and behavioural data alongside test scores, but rarely translate that data into a clear view of which factors actually move outcomes versus which are simply present in the population. Without that view, support resources — tutoring slots, test-prep seats, counselling time — get allocated on intuition rather than evidence.

This project addresses that gap by building an EDA and Power BI semantic model that lets stakeholders:

- monitor average performance across Math, Reading, and Writing at a glance;
- compare outcomes across student subgroups (gender, ethnic group, lunch programme, parental education);
- identify which preparation and household factors carry a measurable effect size, and which do not;
- support data-driven decisions about where to target limited support resources.

Scope note: unlike a sales or operations dataset, this data has no date, term, or semester field, so no time-trend or seasonality analysis is possible. Every finding in this report is a cross-sectional comparison across the 11,040-student snapshot, not a trend over time.

03 Project Objectives

The analysis set out to:

- Profile overall academic performance across Math, Reading, and Writing.
- Quantify the size of the achievement gap associated with lunch-programme enrolment, used as a proxy for economic access.
- Measure the impact of test-preparation completion on scores.
- Compare outcomes by gender and by ethnic group, without over-interpreting demographic correlations as causal.
- Assess the relationship between parental education and student achievement.
- Evaluate household and logistical factors — marital status, first-child status, transport method, sibling count — for material effect.
- Examine study-hour and sport-participation patterns against outcomes.
- Quantify the correlation between the three subject scores to inform whether interventions should be subject-specific or combined.
- Package the findings into an interactive Power BI semantic model and a supporting dashboard for stakeholder use.

04 Dataset Overview

The dataset (Expanded_data_with_more_features.csv) contains 11,040 student records across 15 columns, combining demographic, household, preparation, and outcome fields.

Field summary

Field	Type	Description
Gender	Text	Student gender (female / male)
EthnicGroup	Text	Ethnic group category (group A–E)
ParentEduc	Text	Highest recorded parental education level
LunchType	Text	Standard or free/reduced lunch enrolment
TestPrep	Text	Test-preparation course status (none / completed)
ParentMaritalStatus	Text	Parental marital status
PracticeSport	Text	Frequency of sport practice
IsFirstChild	Text	Whether the student is the first child
NrSiblings	Numeric	Number of siblings
TransportMeans	Text	School bus or private transport
WklyStudyHours	Text	Banded weekly study hours (<5, 5–10, >10)
MathScore	Numeric	Math test score, 0–100
ReadingScore	Numeric	Reading test score, 0–100
WritingScore	Numeric	Writing test score, 0–100

There is no date, academic-term, or school-identifier field, which is why this report does not attempt trend, seasonal, or school-comparison analysis — see the data-quality note in Section 02.

The three outcome fields (MathScore, ReadingScore, WritingScore) are complete or near-complete (WritingScore has a single missing value); the demographic and household fields carry between 2% and 10% missing values, detailed in Section 05.

05 Data Preparation

Preparation followed a standard EDA workflow in Python (pandas), mirrored in the accompanying Excel review:

- Loaded the CSV and reviewed structure with .head(), .info(), and .describe() to confirm row count, column types, and score ranges.
- Dropped the redundant "Unnamed: 0" index column carried over from the source export.
- Ran .isnull().sum() to quantify missing values per column (see table below) — no rows were dropped, since missingness is concentrated in demographic fields not required for every analysis.
- Standardised the WklyStudyHours band labels for consistent grouping.
- Verified numeric fields (scores, sibling count) sit within plausible bounds (0–100 for scores; 0–7 for siblings) with no negative or out-of-range values.
- Confirmed categorical fields (Gender, LunchType, TestPrep, etc.) contain a small, clean set of expected category labels with no free-text contamination.

Missing values by field

Field	Missing	% of 11,040
TransportMeans	1,140	10.3%
EthnicGroup	665	6.0%
TestPrep	659	6.0%
ParentEduc	658	6.0%
NrSiblings	568	5.1%
ParentMaritalStatus	435	3.9%
IsFirstChild	362	3.3%
WklyStudyHours	339	3.1%
PracticeSport	224	2.0%
WritingScore	1	<0.1%

All group-level averages in this report are computed on non-null records for the relevant field, so subgroup sample sizes vary slightly by analysis and are stated alongside each finding.

06 Key Performance Indicators

The Power BI semantic model surfaces five headline KPIs, mirrored in the dashboard's KPI strip (Section 14).

KPI	Value	Why it matters
Student count	11,040	Confirms population size and dataset completeness for scoring fields.
Average Math score	66.7 (σ 15.4)	Headline outcome measure; the widest-spread of the three subjects alongside Writing.
Average Reading score	69.4 (σ 14.7)	Highest-scoring subject on average; tightest spread of the three.
Average Writing score	68.4 (σ 15.4)	Closely tracks Reading (r = 0.95) — see Section 13.
Test-prep completion rate	34.1%	Direct lever with a measurable score lift; low uptake signals room to expand.

These KPIs are deliberately outcome- and lever-focused: three describe where performance currently stands, and the fifth describes the one preparation input stakeholders can most directly influence.

07 Key Business Questions

Framing the analysis as a set of stakeholder questions keeps the findings anchored to decisions rather than to the charts themselves.

Q1. Which student subgroup shows the widest achievement gap, and what does that suggest about equity of access?

Lunch-programme enrolment produces the widest gap of any factor examined: standard-lunch students outscore free/reduced-lunch students by 12.0 Math points, 8.0 Reading points, and 8.9 Writing points. As the closest available proxy for household economic access in this dataset, it points to economic disadvantage as the single largest driver of the achievement gap on file.

Q2. Does completing test preparation meaningfully change outcomes, and is expanding it worth the investment?

Yes. Students who completed test prep scored 4.3 points higher in Math, 6.4 higher in Reading, and 9.4 higher in Writing than those who did not. With only 34.1% of students (with a recorded status) completing it, there is clear headroom to expand access before diminishing returns would be expected to set in.

Q3. Is academic performance uniform across gender, or do strengths differ by subject?

Performance diverges by subject rather than favouring one gender overall: male students average 5.2 points higher in Math, while female students average 6.8 points higher in Reading and 8.7 points higher in Writing. Programme design should treat this as a subject-specific pattern, not a general gender gap.

Q4. How much does parental education level correlate with student outcomes?

Strongly and almost monotonically. Average scores rise from 64.2 (some high school) through 65.5, 67.9, 70.2, 72.4, to 75.2 (master's degree) — an 11.0-point spread from lowest to highest education band, the second-largest gradient in the dataset after lunch type.

Q5. Do household factors such as marital status, birth order, or transport method materially affect scores?

No. Parental marital status, first-child status, and transport method each produce score differences under 3 points across their categories — far smaller than the lunch, test-prep, or parental-education effects. These fields are useful for population description but are not effective targeting levers.

Q6. Does more weekly study time translate into higher scores, and by how much?

Yes, but with a moderate effect size. Students studying more than 10 hours a week average 69.8 points (across subjects) versus 66.4 for those studying under 5 hours — a 3.4-point spread. It is a real but secondary factor relative to lunch type, test prep, and parental education.

Q7. Does sport participation relate to academic performance?

Only modestly. Students who practise sport regularly average about 3 points higher than those who never do (69.2 vs. 66.0), a smaller effect than the academic-support factors above. Sport participation should be framed as a wellbeing initiative, not an academic-lift lever.

Q8. Which ethnic-group segment shows notably different outcomes, and what should stakeholders not infer from that?

Group E (14.3% of students) averages the highest scores across all three subjects (75.8 Math, 74.7 Reading, 73.1 Writing), notably above the other four groups. This should be read as a monitoring signal, not a causal one — ethnic group almost certainly proxies unmeasured socio-economic or household factors already captured more directly by lunch type and parental education, and should not be used on its own as a targeting variable.

Q9. Are the three subject scores independent, or should an intervention aimed at one likely help the others?

Reading and Writing are very tightly linked ($r = 0.95$), while Math correlates more loosely with both ($r \approx 0.81$ – 0.82). Literacy-focused interventions aimed at Reading or Writing should be expected to lift both; Math should be treated as a comparatively distinct skill domain requiring its own resourcing.

Q10. What data is missing that limits deeper trend analysis, such as seasonal or term-over-term patterns?

The dataset has no date, academic-term, or semester field, so month-over-month or seasonal trend analysis — common in retail-style dashboards — is not possible here. Any future data-collection effort aimed at tracking intervention impact over time would need a term or assessment-date field added.

08 Achievement Gap Analysis — Lunch Programme

Lunch-programme enrolment (standard vs. free/reduced) is used throughout this report as the closest available proxy for household economic access, and it produces the single largest score gap of any factor in the dataset.

Lunch type	Students	Avg Math	Avg Reading	Avg Writing
Standard	7,225 (65.4%)	70.83	72.21	71.52
Free / reduced	3,815 (34.6%)	58.84	64.19	62.63
Gap (Standard – Free/reduced)	—	11.99	8.02	8.89

Business impact: this is the widest and most consistent gap examined in the dataset, present across all three subjects, and Math shows the largest absolute difference. It represents the clearest single equity concern the data surfaces.

Recommendation: prioritise supplemental academic support — tutoring blocks, take-home materials, extended library or lab access — for free/reduced-lunch students, and track the gap by subject each reporting cycle to measure whether interventions narrow it.

09 Test-Preparation Impact Analysis

Test-preparation completion is the most directly actionable lever in the dataset — it is a programme stakeholders control, rather than a demographic trait.

Test-prep status	Students	Avg Math	Avg Reading	Avg Writing
Completed	3,538 (34.1%)	69.45	73.65	74.62
None	6,843 (65.9%)	65.16	67.21	65.22
Lift (Completed – None)	—	+4.29	+6.44	+9.40

Business impact: the lift is present in every subject and is largest in Writing (+9.4 points) — yet fewer than 4 in 10 students with a recorded status have completed the course, implying substantial unrealised benefit.

Recommendation: treat test-prep completion as a KPI to actively grow, not just observe. Investigate access barriers (scheduling, cost, awareness) behind the current 34.1% completion rate before assuming demand is the constraint.

10 Demographic Performance — Gender & Ethnic Group

Gender

Gender	Students	Avg Math	Avg Reading	Avg Writing
Female	5,591 (50.6%)	64.13	72.80	72.73
Male	5,449 (49.4%)	69.32	66.00	64.05

Pattern: the two genders are almost evenly split in the dataset (50.6% / 49.4%), and their score differences run in opposite directions by subject rather than one gender leading across the board.

Ethnic group

Group	Students	Share	Avg Math	Avg Reading	Avg Writing
Group C	3,277	31.6%	64.37	68.16	66.77
Group D	2,659	25.6%	67.95	70.50	71.01
Group B	2,111	20.3%	63.52	67.42	65.90
Group E	1,489	14.3%	75.76	74.67	73.09
Group A	839	8.1%	63.61	67.18	65.45

Caution: Group E scores notably higher across all three subjects, but the dataset provides no information on why. Ethnic group most plausibly correlates with — rather than causes — differences already captured by lunch type and parental education, so it should be used for monitoring rather than as a standalone targeting criterion.

11 Household & Parental Factors

Parental education

Parent education	Students	Avg Math	Avg Reading	Avg Writing
Some high school	2,007 (19.3%)	62.81	65.81	63.93
High school	2,010 (19.4%)	64.26	67.04	65.08
Some college	2,375 (22.9%)	66.51	68.91	68.19
Associate's degree	1,994 (19.2%)	68.66	71.44	70.55
Bachelor's degree	1,243 (12.0%)	70.55	73.16	73.45
Master's degree	753 (7.3%)	72.49	76.17	76.86

Gradient: scores rise in step with every increase in parental education level, in all three subjects, with no reversals — one of the cleanest and most consistent patterns in the dataset.

Marital status, first-child status & transport — data-quality note

These three household/logistics fields were examined for completeness, but none showed an effect worth acting on:

- Parental marital status: scores range only 66.1–69.4 across divorced, married, single, and widowed households (widowed is a small group at 219 students) — under 3.5 points end to end.
- First-child status: first children average 0.7–0.9 points higher across subjects than non-first children — a negligible difference.
- Transport method: school-bus and private-transport students differ by less than 0.4 points on every subject.

Data-quality note: these fields do not warrant inclusion in any targeting or resourcing decision on the strength of this dataset alone. They are retained in the model for completeness and population description only.

12 Study Habits & Engagement Analysis

This section stands in for a conventional "monthly trend" analysis, which the dataset cannot support (no date field — see Section 02). Instead, it examines the two behavioural/engagement fields available: weekly study hours and sport participation.

Weekly study hours

Study band	Students	Avg score (3-subject mean)
< 5 hrs/week	2,981 (27.9%)	66.42
5–10 hrs/week	5,797 (54.2%)	68.49
> 10 hrs/week	1,923 (18.0%)	69.78

Trend: each step up in study time is associated with a higher average score, but the total spread (3.4 points from lowest to highest band) is modest next to the lunch, test-prep, or parental-education effects — consistent with diminishing returns beyond a baseline level of study.

Sport participation

Practice frequency	Students	Avg score (3-subject mean)
Never	1,435 (13.3%)	66.02
Sometimes	5,478 (50.9%)	68.04
Regularly	3,903 (36.3%)	69.16

Note: the gap between "never" and "regularly" (about 3.1 points) is similar in size to the study-hours effect, and smaller than the academic-support factors — best framed as a wellbeing indicator rather than an academic-lift programme.

13 Score Correlation Analysis

Pearson correlation across the three subject scores clarifies whether interventions should be designed per subject or combined.

	Math	Reading	Writing
Math	1.00	0.82	0.81
Reading	0.82	1.00	0.95
Writing	0.81	0.95	1.00

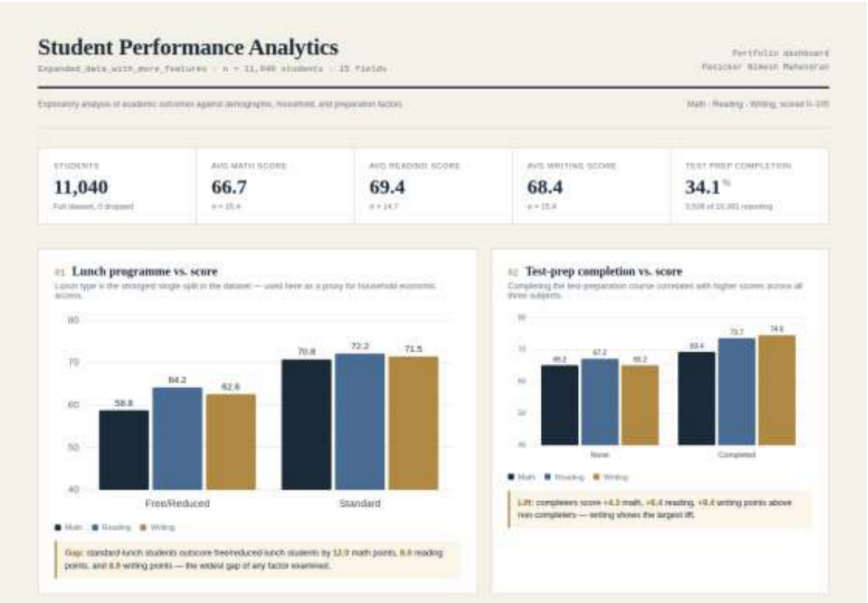
Finding: Reading and Writing move together far more tightly ($r = 0.95$) than either does with Math ($r \approx 0.81$ – 0.82). This is consistent with Reading and Writing sharing an underlying literacy skill set, while Math draws on a more distinct quantitative skill set.

Implication for programme design: a literacy-focused intervention (reading comprehension, writing workshops) is likely to move Reading and Writing scores together, while a Math-specific intervention will likely need to be resourced and measured independently rather than assumed to transfer.

14 Dashboard Overview

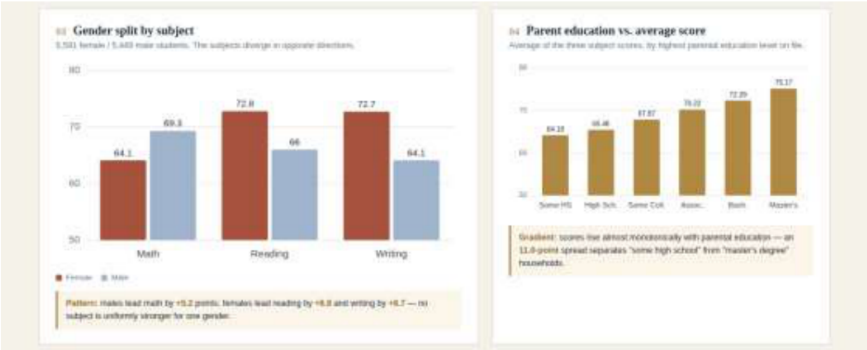
The dashboard below packages the KPIs and comparisons above into a single reference view, built directly from the dataset (no simulated or placeholder figures). It is organised into a KPI strip followed by nine numbered panels, mirroring the section order of this report.

KPI strip & Panels 01–02 — Lunch programme and test-prep impact



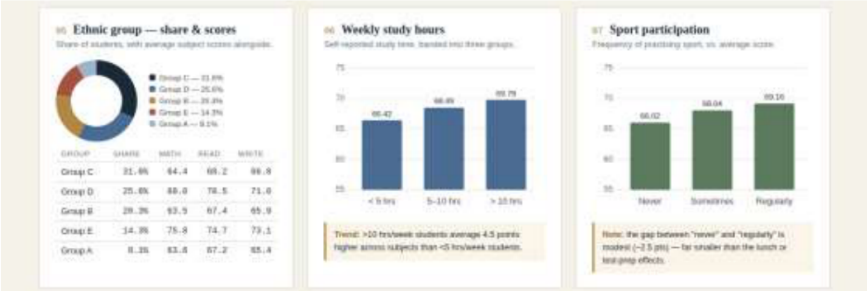
Dashboard, panels 01–02: headline KPIs, lunch-programme gap, and test-preparation lift.

Panels 03–04 — Gender split and parental education gradient



Dashboard, panels 03–04: gender split by subject and the parental-education score gradient.

Panels 05–07 — Ethnic group, study hours, and sport participation



Dashboard, panels 05–07: ethnic-group share and scores, weekly study-hour bands, and sport-practice frequency.

Panels o8–o9 — Score correlation and parental marital status



Dashboard, panels 08–09: the Math/Reading/Writing correlation matrix and the marital-status comparison, included to show the absence of a material effect.

How a stakeholder uses this view: scan the KPI strip for current-state averages, use panels 01–02 to identify the two levers with the largest and most actionable effect sizes (lunch access and test-prep completion), and use panels 03–09 to rule factors in or out before committing support resources.

15 Key Business Insights

Insight 1

Free/reduced-lunch students trail standard-lunch students by up to 12.0 points, the largest gap of any factor in the dataset.

Business impact: This is the clearest single equity concern the data surfaces, and it is large enough to be visible at the individual-student level, not just in aggregate.

Recommended action: Direct supplemental academic support toward free/reduced-lunch students first, and track the gap by subject each reporting cycle.

Insight 2

Only 34.1% of students with a recorded test-prep status have completed the course, despite a 4–9 point score lift.

Business impact: This is the one factor in the dataset that is both actionable and already proven to work — low uptake represents unrealised gains rather than a ceiling effect.

Recommended action: Investigate and remove barriers to test-prep enrolment (scheduling, awareness, cost) and set a completion-rate target for the next cycle.

Insight 3

Parental education level shows an 11.0-point, near-monotonic gradient with student scores across all three subjects.

Business impact: Because this factor cannot be changed by a school programme in the short term, it functions best as a segmentation variable for targeting other interventions (tutoring, test-prep outreach) rather than a lever on its own.

Recommended action: Use parental-education band alongside lunch type to prioritise outreach for support programmes.

Insight 4

Gender differences run in opposite directions by subject: males lead Math by 5.2 points, females lead Reading and Writing by 6.8 and 8.7 points respectively.

Business impact: A single "gender gap" framing would misrepresent the data — the pattern is subject-specific, not a general advantage for one gender.

Recommended action: Design any gender-targeted support (e.g. Math confidence programmes for female students, literacy programmes for male students) around the specific subject where the gap appears.

Insight 5

Reading and Writing scores correlate at $r = 0.95$, far tighter than either does with Math ($r \approx 0.81 - 0.82$).

Business impact: Literacy interventions aimed at one of Reading or Writing are likely to lift both; Math requires independent resourcing and cannot be assumed to improve as a side effect.

Recommended action: Bundle Reading and Writing support into a combined literacy track, and fund Math support as a separate track with its own goals.

Insight 6

Group E students (14.3% of the population) average 6–12 points higher than the other four ethnic groups across all subjects.

Business impact: Without further data, this is a monitoring signal rather than actionable insight — it likely reflects socio-economic factors already captured by lunch type and parental education.

Recommended action: Do not use ethnic group as a standalone targeting variable; monitor it for equity reporting purposes alongside the more directly actionable factors.

Insight 7

Household composition factors — marital status, first-child status, transport method — each show under 3 points of score difference across categories.

Business impact: These fields add descriptive completeness to the dataset but carry no meaningful signal for resource allocation.

Recommended action: Exclude these fields from any targeting model; retain them only for population reporting.

Insight 8

Weekly study hours show a modest, positive, and diminishing-returns relationship with scores (3.4-point spread from <5 hrs to >10 hrs/week).

Business impact: Study time matters, but its effect size is secondary to structural factors like lunch access and test preparation.

Recommended action: Encourage baseline study habits (reaching the 5–10 hr/week band) rather than promoting open-ended additional hours as a primary lever.

Insight 9

Sport participation shows a similarly modest positive relationship (about 3 points, never vs. regularly) to overall scores.

Business impact: The effect size is comparable to study hours and smaller than the socio-economic and preparation factors — best treated as a wellbeing benefit rather than an academic strategy.

Recommended action: Continue supporting sport access for its wellbeing value, without positioning it as an academic-outcomes programme.

16 Business Recommendations

Ranked by the size of the effect each is connected to, largest first:

- 1. Close the lunch-programme gap first.**
It is the single largest and most consistent effect in the data (up to 12.0 points). Direct tutoring, materials, and support resources toward free/reduced-lunch students as the top priority.
- 2. Grow test-preparation completion from its current 34.1%.**
It is the most directly actionable lever with a proven 4–9 point lift. Set a completion-rate KPI and investigate access barriers before assuming low demand.
- 3. Use parental-education band as a segmentation variable, not a lever.**
The 11.0-point gradient is real but not something a programme can change directly — use it to prioritise which students receive outreach for the levers above.
- 4. Bundle Reading and Writing support; resource Math independently.**
Their 0.95 correlation means literacy interventions likely serve both; Math's lower correlation (≈0.81) means it needs its own programme and its own success metric.
- 5. Design subject-specific, not blanket, gender support.**
Since males lead Math and females lead Reading/Writing, any gender-targeted programme should name the specific subject gap it is addressing.
- 6. Monitor, but do not target on, ethnic group or household composition.**
These fields either likely proxy other factors (ethnic group) or show no material effect (marital status, first-child status, transport). Keep them in equity reporting, not in resource-allocation rules.
- 7. Treat study hours and sport participation as secondary, supportive factors.**
Both show real but modest effects. Encourage healthy baselines in each without over-promising academic impact.
- 8. Add a date or term field to future data collection.**
The current dataset cannot support trend or intervention-impact-over-time analysis. Capturing an assessment date or academic term would let future reporting measure whether these recommendations actually move the gaps identified here.

17 Conclusion

Across 11,040 students, this analysis finds that academic outcomes are shaped far more strongly by economic access and test-preparation completion than by household composition or demographic category on their own. The lunch-programme gap (up to 12.0 points) and the test-prep lift (4–9 points) are the two clearest, most actionable findings in the dataset — one describing an equity concern that needs support resourcing, the other describing an under-used lever that is already proven to work.

Parental education adds a further, consistent 11.0-point gradient that is useful for segmentation even though it cannot be changed directly by a school programme. Gender, ethnic group, study hours, and sport participation each add real but smaller nuance, while marital status, first-child status, and transport method contribute essentially nothing to an allocation decision.

The main risk in this dataset is over-interpreting demographic correlations — particularly ethnic group — as causal drivers rather than proxies for the economic and preparation factors already captured more directly elsewhere. The Power BI semantic model and dashboard built for this project are designed to keep that distinction visible: KPIs and levers are surfaced first, with demographic breakdowns available for monitoring rather than targeting.

With a small addition to future data collection — a date or term field — the same model could extend from a cross-sectional snapshot to a genuine before/after measurement of whether the recommendations in Section 16 actually move the gaps identified here.

18 Portfolio Project Summary

Project Name

Student Performance Analytics

Tools Used

Microsoft Excel, Python (pandas, seaborn, matplotlib), Power BI Semantic Modelling

Business Objective

Identify which demographic, household, and preparation factors are most strongly associated with student achievement across Math, Reading, and Writing, and separate actionable levers from purely descriptive traits.

Key Analysis

Exploratory data analysis and missing-value review in Python; grouped comparisons across lunch type, test-prep status, gender, ethnic group, parental education, household composition, study hours, and sport participation; a Pearson correlation analysis across the three subject scores; and a Power BI semantic model with calculated columns and DAX measures supporting an interactive dashboard.

Key Findings

Lunch-programme enrolment and test-preparation completion are the two strongest, most actionable factors, with parental education adding a further consistent gradient. Household composition and transport method show no material effect, and demographic categories such as ethnic group should be treated as monitoring signals rather than targeting variables.

Business Value

The dashboard gives stakeholders a single reference view to prioritise support resources by effect size rather than intuition, and flags where demographic patterns require caution rather than direct action.

19 Skills Demonstrated

- Data Cleaning & Preparation (Python/pandas)
- Missing-Value Analysis
- Exploratory Data Analysis (EDA)
- Statistical Correlation Analysis
- Microsoft Excel
- Power BI Semantic Modelling
- DAX Measures & Calculated Columns
- Data Visualisation
- Dashboard Development
- KPI Development
- Business Analysis & Insight Framing
- Data Storytelling
- Equity & Demographic-Bias Awareness in Analysis
- Data-Driven Recommendation Writing